

A
Major Project Report on
**NLP Based Resume Analysis
and Ranking System**

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for the degree of
BACHELOR OF ENGINEERING
IN
Computer Science & Engineering
Artificial Intelligence & Machine Learning

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ABSTRACT

The recruitment process often involves analyzing thousands of resumes, making manual screening time-consuming and biased. To address this, the project introduces a Natural Language Processing (NLP)-Based Resume Analysis and Ranking System that uses NLP, Machine Learning (ML), and Deep Learning (DL) techniques to automate candidate evaluation and improve hiring accuracy.

The system enables students to upload their resumes, compare them with job descriptions, and receive insights such as Applicant Tracking System (ATS) score, success rate, strengths, weaknesses, and skill gaps. It also provides personalized recommendations, course suggestions, and mock interviews to enhance employability. For Human Resource (HR) professionals, the system supports job posting, bulk resume upload, automatic candidate ranking, and analytical dashboards for better decision-making.

Its core intelligence combines Term Frequency–Inverse Document Frequency (TF-IDF) and cosine similarity for keyword matching, Sentence Transformers (MiniLM) and Facebook AI Similarity Search (FAISS) for semantic similarity, Bidirectional and Auto-Regressive Transformer for Natural Language Inference (BART NLI) for skill entailment detection, Job-domain Bidirectional Encoder Representations from Transformers (JobBERT) for entity extraction, and Long Short-Term Memory (LSTM) for career path prediction. Operating on local models ensures data privacy, real-time performance, and cost efficiency.

The project aligns with the United Nations Sustainable Development Goals (SDGs) — SDG 4 (Quality Education), SDG 8 (Decent Work and Economic Growth), and SDG 9 (Industry, Innovation, and Infrastructure) by promoting fair, transparent, and intelligent recruitment.

Keywords: NLP, Machine Learning, Deep Learning, Resume Analysis, Candidate Ranking, ATS Score, Semantic Similarity, Term Frequency–Inverse Document Frequency, Sentence Transformers, Facebook AI Similarity Search, Bidirectional and Auto-Regressive Transformer for Natural Language Inference, Job-domain Bidirectional Encoder Representations from Transformers, Long Short-Term Memory.

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ABBREVIATIONS

ATS	Applicant Tracking System
BART	Bidirectional and Auto-Regressive Transformer
BERT	Bidirectional Encoder Representations from Transformers
CORS	Cross-Origin Resource Sharing
CSV	Comma-Separated Values
DFD	Data-Flow Diagram
FAISS	Facebook AI Similarity Search
JWT	JSON Web Token
LSTM	Long Short-Term Memory
MiniLM	Minimal BERT (distilled transformer)
ML	Machine Learning
NER	Named Entity Recognition
NLP	Natural Language Processing
NLI	Natural Language Inference
ORM	Object-Relational Mapping
RoBERTa	Robustly Optimized BERT Pre-Training Approach
SHAP	SHapley Additive exPlanations
SQL	Structured Query Language
SQLite	Structured Query Language Lite
TF-IDF	Term Frequency–Inverse Document Frequency
UUID	Universally Unique Identifier
Uvicorn	Ultra-fast ASGI server

Chapter 1

Introduction

In today's competitive employment landscape, recruiters receive hundreds or even thousands of résumés for a single job posting [1][2]. Manual screening is not only time-consuming but also subject to human bias and inconsistency [5]. Traditional applicant tracking systems partially automate the process via keyword searches, yet they fail to capture contextual meaning or overall suitability [3][15].

To overcome these limitations, this project introduces an NLP-based résumé analysis and ranking system that applies NLP, ML, and DL algorithms to analyse résumés, compare them with job descriptions, and generate explainable, data-driven insights [2][3][9]. The system bridges the gap between job seekers and recruiters by enabling fair, transparent, and efficient candidate evaluation [5][11].

The proposed model performs multi-stage analysis that mimics human interpretation while ensuring computational efficiency. It employs TF-IDF with cosine similarity for keyword-level comparison [15], Sentence Transformer embeddings (MiniLM) and FAISS for contextual semantic matching [3], BART NLI for skill-entailment detection [2], JobBERT for skill and entity extraction [10][14], and an LSTM network for career-path prediction [2]. These models collectively deliver an accurate ATS score, highlight strengths and weaknesses, identify skill gaps, and generate personalised recommendations—all without relying on external APIs, thereby preserving data privacy and minimising operational cost [6][13].

From the student perspective, the system allows users to upload résumés, select a target job description, and receive a detailed analysis that includes ATS score, keyword match percentage, success rate, matched and missing skills, improvement suggestions, and recommended online courses [9][12]. Students can also participate in mock interviews and track their application progress through a visual analytics dashboard [7].

From the HR perspective, recruiters can create and manage job postings, upload résumés in bulk, automatically rank candidates, view comparative analyses of applicant strengths and weaknesses, and schedule interviews within the same interface [1][8]. The ranking mechanism combines semantic similarity, keyword matching, and experience relevance to ensure an objective and explainable evaluation process [3][9][11].

By integrating these features within a unified architecture built on FastAPI (backend), React + Chakra UI (frontend), and SQLite (database), the system delivers a seamless and secure workflow for both users [6]. The result is a scalable, transparent, and intelligent ATS platform that enhances recruitment quality while empowering candidates to continuously improve their professional profiles [5][11].

This report presents a detailed study of the NLP-Based Résumé Analysis and Ranking System, outlining its purpose, design, and implementation. It covers a literature review, problem statement, objectives, scope, and the proposed system architecture with implementation details. The report also includes the experimental setup, project plan, and expected outcomes, highlighting how the system enhances recruitment efficiency, accuracy, and transparency through intelligent automation.

Chapter 2

Literature Survey

History

The process of recruitment and resume screening has evolved significantly over the past few decades, moving from entirely manual evaluation to semi-automated and now AI-driven systems. In the early stages of recruitment technology (1980s–2000s), most organizations relied on manual resume screening. This approach was time-consuming, inconsistent, and susceptible to human bias, especially in large-scale hiring.

With the rise of digital recruitment portals such as Monster and Naukri in the early 2000s, the first generation of Applicant Tracking Systems (ATS) emerged. These early systems primarily relied on rule-based keyword matching meaning scanning resumes for specific words or phrases that matched job descriptions. While this automation reduced manual workload, it often led to inaccurate results because it could not understand context, synonyms, or semantic relationships between skills and job requirements.

By the 2010s, advancements in Machine Learning (ML) began to influence recruitment automation. Techniques such as TF-IDF (Term Frequency–Inverse Document Frequency) and cosine similarity were introduced to measure the importance and similarity of words across resumes and job descriptions. These methods improved precision by considering term frequency and relevance but still struggled with understanding contextual meaning.

The next major shift came with the advent of Natural Language Processing (NLP) and Deep Learning (DL) models. Pre-trained transformer architectures such as BERT (Bidirectional Encoder Representations from Transformers), RoBERTa, and MiniLM revolutionized text understanding by enabling semantic and contextual comprehension of language. These models allowed modern ATS systems to move beyond keyword-based filtering to context-aware resume analysis, significantly improving matching accuracy.

In recent years, AI-driven ATS platforms have incorporated hybrid techniques combining traditional NLP (TF-IDF, regex) and transformer-based deep learning models for semantic similarity, entity recognition, and zero-shot classification. This integration has led to more reliable candidate ranking, fair evaluation, and improved hiring efficiency.

The NLP-Based Resume Analysis and Ranking System proposed in this project is the result of this technological evolution. It builds on these historical developments by combining TF-IDF,

Sentence Transformers, FAISS, BART NLI, JobBERT, and LSTM models into one unified framework, achieving both explainability and accuracy without dependence on external cloud APIs.

Literature Review

[1] Machine Learning-Based Human Resource Management Information Retrieval and Classification Algorithm

Year and Authors: 2024, Wen Li and XiuKao Zhou

This paper applies machine learning techniques for information retrieval and document classification to streamline HR data management. It improves the efficiency of candidate search and filtering using ranking-based retrieval models, a concept reflected in this project's FAISS-powered semantic search engine for job–resume matching.

[2] Automated Resume Parsing: A Natural Language Processing Approach

Year and Authors: 2023, Sougandh, Thatavarthi Giri, Nithish Sagar Reddy, and Meena Belwal

This paper focuses on automating the extraction of structured information such as skills, education, and experience from unstructured resume documents. It employs NLP libraries like spaCy, PyMuPDF, and Transformer models to identify key sections and entities. The work highlights how modern parsing systems can replace manual extraction by leveraging language models for entity recognition and document segmentation.

[3] VacancySBERT: Representation of Titles and Skills for Semantic Similarity Search

Year and Authors: 2023, Maiia Bocharova, Eugene Malakhov, and Vitaliy Mezhuyev

This research proposes Sentence-BERT (SBERT) as a method for embedding job titles and skills to measure semantic similarity between resumes and job descriptions. The model ensures that candidate–job matching goes beyond keyword overlap by capturing contextual meaning. This aligns closely with the proposed system's semantic similarity component based on Sentence Transformers (MiniLM).

[4] A Machine Learning Approach to Detecting Fraudulent Job Types

Year and Authors: 2023, Marcel Naudé, Kolawole John Adebayo, and Rohan Nanda

This work addresses the issue of fraudulent or exaggerated resume content by employing AI-based anomaly detection. Using outlier detection and data consistency analysis, it identifies suspicious applications or false information. This contributes to secure and reliable recruitment automation, complementing the proposed system's validation and verification logic.

[5] Enhancing Diversity and Reducing Bias in Recruitment through AI: A Review of Strategies and Challenges

Year and Authors: 2023, Ramakrishnan Vivek

This review discusses how AI and explainable machine learning can help minimize unconscious bias in recruitment. It emphasizes fairness, transparency, and interpretability as core values of ethical HR technology. These principles directly influence the design philosophy of the proposed NLP-based ranking system, which ensures explainable and equitable candidate assessment.

[6] Secured Recruitment Management System with Data Retention Approach

Year and Authors: 2023, Mashitah Mohd Tobroni and Isredza Rahmi A. Hamid

This study focuses on implementing data security in recruitment systems, emphasizing JWT authentication, role-based access control, encryption, and CSRF protection. The proposed system adopts a similar security model through FastAPI's JWT and role-based authentication to ensure privacy of sensitive candidate and recruiter information.

[7] Illuminating HR Insights: Unraveling Patterns through Data Analytics and Visualization

Year and Authors: 2023, Akshata Upadhye

This paper explores data analytics and visualization techniques to provide actionable insights for HR professionals. It introduces dashboards for analyzing applicant trends, skill gaps, and hiring patterns. The proposed system extends this concept by integrating interactive analytics dashboards (using Recharts and Chakra UI) for both students and HR users to monitor performance and recruitment metrics.

[8] Job Candidate Rank Approach Using Machine Learning Techniques

Year and Authors: 2021, Lamiaa Mostafa and Sara Beshir

This paper introduces machine learning models for candidate scoring and ranking based on resume features. It applies methods like vector similarity, regression, and classification to generate unbiased, data-driven rankings. The study establishes the foundation for systems that use ML-based ranking pipelines similar to the proposed project's RankNet-style candidate evaluation mechanism.

The reviewed research papers collectively highlight major advancements in NLP and ML for recruitment automation. Their key aspects including domain, algorithm, dataset, and identified research gaps are summarized in a table mentioned below.

Year	Title	Domain	Algorithm	Dataset	Gap
2024	Machine Learning-Based HR Management Information Retrieval and Classification Algorithm	HR Data Retrieval & Classification	Information retrieval (IR) models, ML classifiers, ranking-based retrieval	HR document repository	Emphasizes document retrieval; does not perform ATS or candidate–job matching.
2023	Automated Resume Parsing: A Natural Language Processing Approach	Resume Parsing, Information Extraction	spaCy NER, PyMuPDF for text extraction, Transformer models for entity recognition	Custom dataset of PDF/DOCX resumes	Focuses only on parsing structured info (skills, education, experience); lacks candidate ranking or ATS scoring.
2023	VacancySBERT: Representation of Titles and Skills for Semantic Similarity Search	Semantic Matching in Recruitment	Sentence-BERT (SBERT) for embeddings, cosine similarity for semantic search	Proprietary job–resume corpus	Addresses semantic similarity but does not integrate ranking or skill gap detection.
2023	A Machine Learning Approach to Detecting Fraudulent Job Types	Resume Verification, Fraud Detection	Outlier detection, anomaly detection, AI-based consistency checks	Online job postings dataset	Detects fraud but not suitable for resume–job semantic analysis or scoring.

2023	Enhancing Diversity and Reducing Bias in Recruitment through AI	Fairness & Ethical AI in HR	Explainable AI, bias detection, fairness-aware ML models	Literature-based study	Focuses on ethics and transparency but lacks technical implementation for resume analysis.
2023	Secured Recruitment Management System with Data Retention Approach	Recruitment System Security	JWT authentication, RBAC, encryption, CSRF protection	Prototype recruitment platform	Focuses only on system security, not AI/NLP-driven analysis or ranking.
2023	Illuminating HR Insights: Unraveling Patterns through Data Analytics and Visualization	HR Analytics & Visualization	Data analytics, visualization using charts and dashboards	HR data analytics dataset	Provides visualization but lacks AI-driven scoring, NLP, or predictive modeling.
2021	Job Candidate Rank Approach Using Machine Learning Techniques	Candidate Ranking	Regression and classification models; vector similarity for scoring	Collected candidate resumes and job posts	ML-based ranking without NLP context understanding; lacks semantic embeddings.

Table 2.1 Summary of Literature Review

Chapter 3

Limitations of Existing Systems

The traditional recruitment and resume screening process is largely manual or relies on basic automation techniques that fail to deliver accuracy, speed, and fairness. While various Applicant Tracking Systems (ATS) and HR analytics platforms exist, they exhibit several limitations in terms of intelligence, feedback, and usability. The key limitations of the existing systems are summarized below.

1. Manual Screening and Subjectivity

Most organizations still rely on manual resume screening, which is time-consuming and inconsistent. Human evaluators may introduce biases and errors in candidate selection, reducing fairness and efficiency.

2. Keyword Dependency

Conventional ATS tools depend solely on keyword matching rather than understanding meaning. Candidates who use synonyms or different phrasing for the same skill (e.g., “ML” vs. “Machine Learning”) are often overlooked, resulting in inaccurate shortlisting.

3. Lack of Semantic Understanding

Existing systems fail to capture the contextual and semantic similarity between a candidate’s resume and job description. Without deep NLP models or embeddings, these systems cannot evaluate true relevance beyond word-level overlap.

4. Limited Feedback to Applicants

Most recruitment systems only display a status update (such as “shortlisted” or “rejected”) without offering detailed insights, such as ATS score, strengths, weaknesses, or improvement suggestions, leaving applicants unaware of how to enhance their profiles.

5. Absence of Skill Gap Analysis and Recommendations

Current solutions do not provide skill gap detection or course recommendations to help candidates improve missing competencies, restricting opportunities for self-assessment and upskilling.

6. No Integration of Mock Interviews or Success Prediction

Existing tools lack mock interview features and success rate prediction models that can guide applicants in preparing for real interviews or estimating their probability of selection based on resume–job fit.

7. Inadequate Candidate Ranking Mechanisms

Many platforms use rule-based filters or manual sorting instead of employing data-driven ranking models such as TF-IDF scoring, semantic similarity, or entailment reasoning. This limits their ability to prioritize the most suitable candidates effectively.

8. Security and Privacy Concerns

Numerous existing systems do not enforce strong security protocols like JWT authentication, role-based access control, or data encryption, exposing sensitive personal information to potential misuse.

9. Lack of a Unified Platform for HR and Students

Most recruitment tools are fragmented, with separate platforms for candidates, HR managers, and analytics. This creates integration issues and prevents a seamless end-to-end workflow for resume uploading, job posting, ranking, and feedback management.

Chapter 4

Problem Statement, Objectives and Scope

Problem Statement

The traditional recruitment process is slow, biased, and inefficient due to manual resume screening and keyword-based Applicant Tracking Systems (ATS). These systems fail to understand semantic relationships between job descriptions and resumes, often filtering out suitable candidates.

Moreover, existing tools provide little to no feedback, lack skill gap analysis, and do not offer predictive insights or mock interview support for applicants. Recruiters, on the other hand, struggle to rank candidates objectively and securely manage large-scale applications.

Therefore, there is a need for an intelligent, secure, and NLP-based system that can automate resume evaluation, compute ATS and semantic similarity scores, rank candidates objectively, and provide meaningful feedback to enhance both recruitment efficiency and candidate preparedness.

Objectives

The primary objective of this project is to design and develop an AI-powered recruitment system that automates resume analysis and candidate ranking using Natural Language Processing (NLP) and Machine Learning (ML). The system aims to bridge the gap between candidates and employers by providing intelligent, fair, and data-driven hiring decisions.

The specific objectives of this project are as follows:

1. Automate Resume Evaluation:

To automatically analyze and extract key details such as skills, education, and experience from resumes using NLP and deep learning models.

2. Enhance Candidate–Job Matching:

To compute similarity between resumes and job descriptions using TF-IDF and Sentence Transformer embeddings for accurate relevance scoring.

3. Generate Explainable ATS Scores:

To provide a transparent and data-driven Applicant Tracking Score based on keyword relevance, semantic matching, and skill alignment.

4. Identify Skill Gaps and Recommend Improvements:

To detect missing skills and suggest relevant online courses or learning resources for upskilling.

5. Support Mock Interviews and Success Prediction:

To assist students in improving employability by generating interview questions and estimating success probability for target job roles.

6. Enable Intelligent Candidate Ranking for HR:

To rank applicants objectively using a hybrid scoring model based on semantic similarity, experience, and qualifications.

7. Ensure Data Security and Privacy:

To implement JWT authentication, role-based access control, and secure data handling for both candidates and recruiters.

8. Provide Unified Platform for Students and HR:

To integrate all functionalities—resume analysis, ranking, application tracking, and insights—into a single user-friendly web platform.

Scope

The scope of the NLP-Based Resume Analysis and Ranking System extends across the complete recruitment lifecycle for both candidates and recruiters. The system aims to create an intelligent, automated, and transparent platform that enhances employability and streamlines hiring.

For Students and Job Seekers

- Upload, analyse, and improve resumes using AI-driven feedback.
- Receive ATS scores, success probability, and skill gap insights.
- Access course recommendations to strengthen missing competencies.
- Participate in mock interviews to boost confidence and performance.
- Track job applications and manage activity history through analytics dashboards.

For HR Professionals

- Create and manage job postings with detailed requirements.

- Upload and analyse resumes in bulk for ATS scoring and ranking.
- View candidate comparisons based on skills, experience, and education.
- Manage interview scheduling and track recruitment metrics via dashboards.

System-Level Scope

- Implements NLP, ML, and DL algorithms (TF-IDF, Sentence Transformers, FAISS, BART NLI, JobBERT, LSTM) for semantic analysis and ranking.
- Ensures end-to-end data security through JWT authentication, RBAC, and encryption.
- Operates as a full-stack web application using FastAPI (backend) and React with Chakra UI (frontend).
- Provides scalability for integration of future AI modules such as predictive hiring analytics, chatbots, and video interview analysis.

Chapter 5

Proposed System

Block Diagram

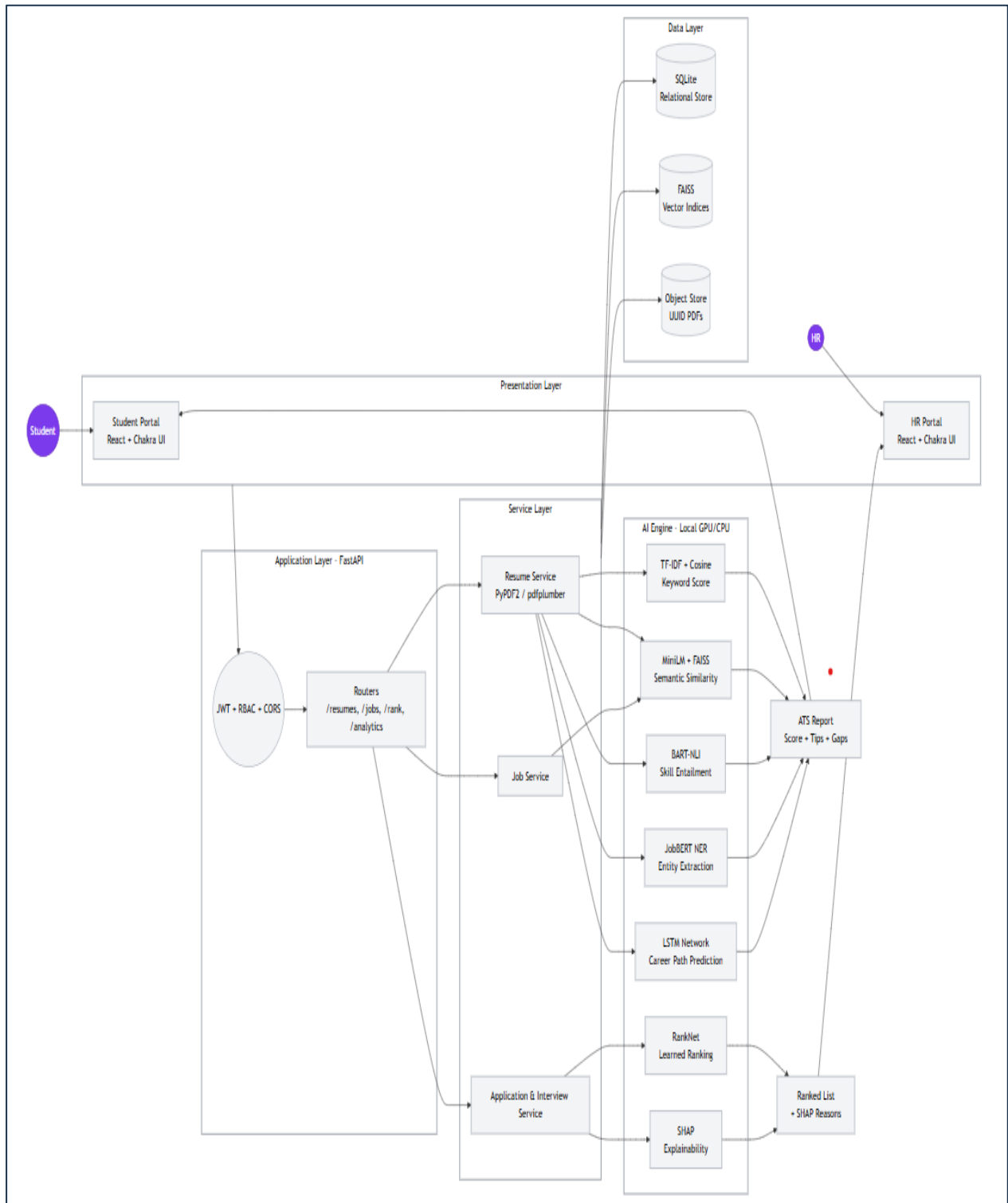


Figure 5.1 Architectural Diagram of Proposed System

Description of Block Diagram

User Layer

The diagram begins with two external actors: Student and HR. Each interacts with a dedicated React + Chakra UI portal [6][7]. Students upload résumés, select target jobs, and receive an ATS report that contains the score, strengths, weaknesses, and skill-gap advice [2][3][9]. HR staff create job postings, bulk-upload résumés, and view a ranked short-list accompanied by SHAP-based explanations for every candidate [3][11].

Application Layer

Both portals communicate with a single FastAPI backend secured by JWT + RBAC + CORS [6]. This layer exposes REST endpoints under /resumes, /jobs, /rank, and /analytics [1]. JWT access tokens are rotated, roles are enforced (student vs HR), and CORS policies restrict cross-origin access [6].

Service Layer

Three stateless services orchestrate business logic. The Resume Service ingests PDF/DOCX files, extracts clean text with PyPDF2 or pdfplumber, and hands the text to the AI Engine [2]. The Job Service manages job descriptions and automatically extracts skills, degrees, and companies via JobBERT NER [10][14]. The Application & Interview Service handles apply actions, status changes, interview scheduling, and final ranking generation [8].

AI Engine (Local GPU/CPU)

The proposed system begins by taking a resume file and a target job description as input. The text content is first extracted using *PyPDF2* and then cleaned and pre-processed using *regular expressions (regex)* and *NLTK* functions such as tokenization and stopword removal[2]. Next, JobBERT (Job-domain Bidirectional Encoder Representations from Transformers) is applied to perform Named Entity Recognition (NER) and extract key entities such as skills, education, and professional experience[10][14].

To compute the similarity between a candidate's resume and a job description, Term Frequency–Inverse Document Frequency (TF-IDF) along with cosine similarity is used for keyword-based matching[15]. To capture deeper semantic relationships beyond keyword overlap, Sentence Transformers (MiniLM) are employed to generate contextual embeddings, and Facebook AI Similarity Search (FAISS) is used for fast vector-based similarity retrieval[3]. For skill entailment detection, the Bidirectional and Auto-Regressive Transformer for Natural Language Inference (BART NLI) model is utilized to check whether the candidate's skills semantically satisfy the job's requirements[2]. Additionally, Long Short-

Term Memory (LSTM) networks are used for career path prediction, helping identify potential future roles based on previous experience patterns [2].

Finally, multiple scores such as keyword match, semantic similarity, skill entailment, experience, and education are aggregated into a single Applicant Tracking System (ATS) score. The system then generates a detailed output containing the candidate's ranking, matched and missing skills, success probability, and personalized feedback.

Data Layer

SQLite holds relational data (users, jobs, applications, interviews, logs) [6]. FAISS vector indices cache MiniLM/MPNet embeddings for millisecond similarity look-ups [3]. A local object store keeps immutable résumé PDFs addressed by UUID [6].

Data Flow Diagram

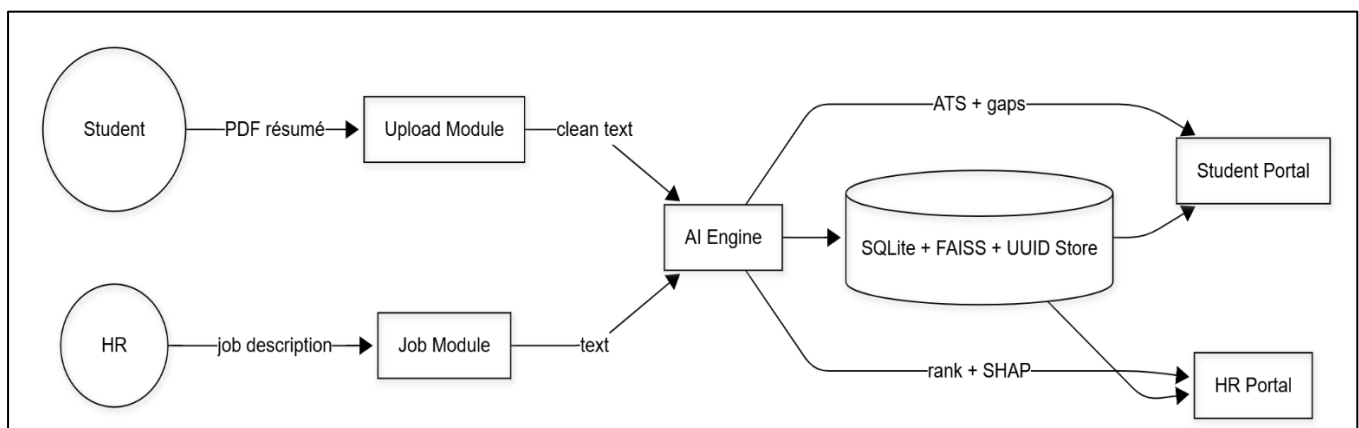


Figure 5.2 Level 0 Diagram

In figure 5.2, the Level-0 diagram situates the system within its external environment. Two primary actors, Student and HR, provide inputs: a PDF résumé and a textual job description, respectively. These data streams converge into a single central process labelled “AI Engine,” which encapsulates all internal intelligence. The engine produces two distinct outputs: an ATS report (containing score, skill gaps, and improvement tips) routed to the Student Portal, and a ranked candidate list accompanied by SHAP explanations delivered to the HR Portal. All persistent artefacts—relational records, vector indices, and UUID-addressed PDF files—are stored in the central “SQLite + FAISS + UUID Store” data store. By concealing algorithmic detail, the diagram satisfies DFD-0 conventions and clearly demarcates the system boundary.

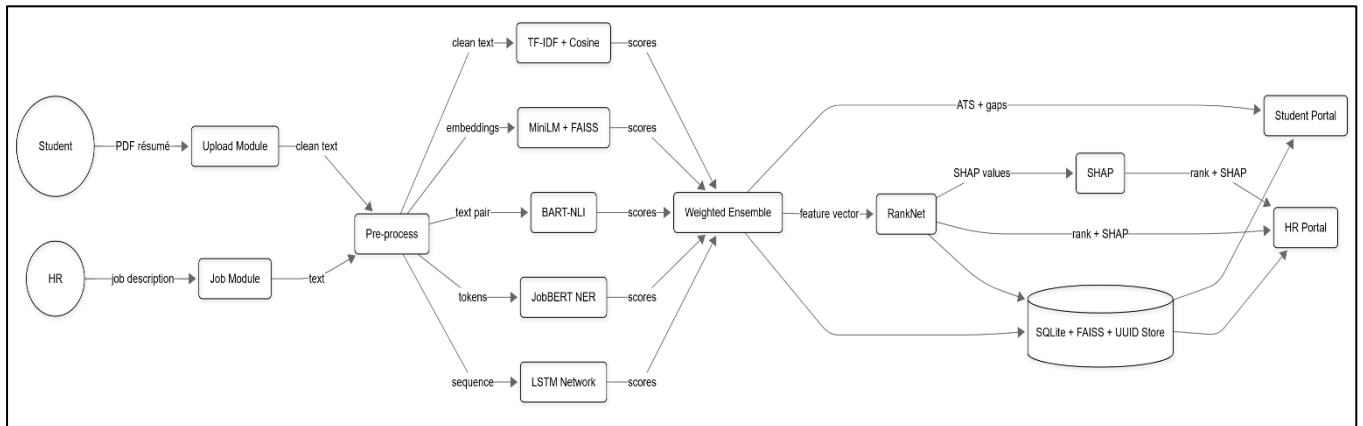


Figure 5.3 Level 1 Diagram

In figure 5.3, to expose internal logic without sacrificing formal clarity, the Level-0 “AI Engine” process is decomposed into six concurrent sub-processes. Pre-processing normalises raw text into tokens suitable for downstream analysis. Concurrently, five specialised modules generate independent similarity or entailment scores: TF-IDF + Cosine for keyword overlap, MiniLM + FAISS for dense semantic similarity, BART-NLI for zero-shot skill-gap detection, JobBERT NER for entity extraction, and an LSTM Network for career-path prediction. Their outputs feed a “Weighted Ensemble” process that computes the final 0-100 ATS value and assembles a six-dimensional feature vector. This vector is forwarded to “RankNet,” a learned pairwise ranker, whose logits are subsequently explained by “SHAP” to yield per-candidate attribution bars. Resultant rankings and explanations are persisted in the central store and relayed to the respective portals.

Key Components of the Proposed System

The proposed framework consists of several interconnected components that work together to achieve accurate and explainable resume evaluation. Following are the components used in the project:

1. Resume Parsing Module

This module is responsible for extracting and structuring data from uploaded resumes in PDF or DOCX format. Using libraries such as PyPDF2 and pdfplumber, the textual content is extracted and then processed using JobBERT for Named Entity Recognition (NER) to identify key entities such as skills, education, work experience, and company names. It converts

unstructured resume text into structured data, which is then stored in the database for further analysis. This step forms the foundation for accurate downstream processing.

2. NLP Analysis Engine

The NLP Analysis Engine forms the core intelligence of the system. It employs a hybrid combination of traditional and deep learning-based Natural Language Processing models to evaluate resumes against job descriptions. TF-IDF (Term Frequency–Inverse Document Frequency) and cosine similarity are used for keyword-level matching, while Sentence Transformers (MiniLM) and FAISS (Facebook AI Similarity Search) handle semantic similarity. Additionally, BART NLI (Bidirectional and Auto-Regressive Transformer – Natural Language Inference) is used for skill entailment detection, and JobBERT aids in refining entity-level accuracy. This module ensures both lexical and contextual understanding of resume and job content.

3. Scoring and Ranking Module

This module aggregates results from the NLP Analysis Engine to produce an overall Applicant Tracking System (ATS) score. It combines multiple evaluation factors such as keyword match percentage, semantic similarity score, skill match ratio, experience relevance, and education alignment. Based on these parameters, candidates are ranked and assigned a success probability score. For HR users, it provides ranked candidate lists with reasoning, while for students, it highlights strengths, weaknesses, and matched versus missing skills.

4. Recommendation Module

The Recommendation Module identifies skill gaps by comparing a student's extracted skills with the requirements mentioned in job descriptions. Using ontology-based skill mapping, it expands related concepts (e.g., “React” → “JavaScript”) to ensure comprehensive detection. Once missing skills are identified, the system suggests relevant online courses or certifications to help students enhance their profiles and improve job readiness.

5. Mock Interview Module

This module supports student preparation by offering AI-generated mock interviews. It evaluates user responses based on predefined question banks, soft skill indicators, and technical knowledge areas relevant to the applied job. Feedback is generated in terms of

communication skills, confidence, and subject understanding, helping users strengthen their interview performance before real assessments.

6. Dashboard and Analytics Module

The Dashboard and Analytics Module provides a unified interface for both students and HR users. Students can view ATS scores, resume feedback, skill gap charts, and application statistics, while HRs can analyze candidate pipelines, recruitment efficiency, and source effectiveness. It utilizes Chakra UI and Recharts for interactive data visualization and ensures real-time tracking of performance metrics and hiring analytics.

7. Security and Access Control Module

To maintain system integrity and data privacy, this module incorporates JWT (JSON Web Token)-based authentication and role-based access control (RBAC). It ensures that only authorized users can access their respective modules — students manage resumes and analytics, while HRs handle job postings and applications. Additional mechanisms such as rate limiting, CORS handling, and encrypted data storage further enhance security and compliance.

Chapter 6

Experimental Setup

Software Requirements:

- Operating System: Windows 10 / 11 (64-bit) or Ubuntu 22.04 LTS
- Backend Framework: FastAPI (Python-based high-performance web framework)
- Frontend Technologies: React 18+, Chakra UI, Vite, Axios, React Router DOM
- Database: SQLite (lightweight relational database)
- File Handling: PyPDF2 / pdfplumber for document parsing, local file storage system
- Authentication: JWT (JSON Web Tokens) using python-jose and passlib (bcrypt)
- API Documentation: Auto-generated Swagger UI from FastAPI
- Visualization Libraries: recharts / chart.js for analytics dashboards
- Logging and Rate Limiting: Python logging module and slowapi
- Environment Management: python-dotenv

Hardware Requirements:

- Processor: Intel Core i5 (8th Gen or above) / AMD Ryzen 5 or higher
- RAM: Minimum 8 GB (Recommended 16 GB for model execution)
- Storage: Minimum 10 GB free space
- GPU (Optional): NVIDIA GPU with CUDA support for faster DL inference
- Display: 1080p monitor with 60Hz refresh rate

Operating Environment:

- Web Browser: Google Chrome / Microsoft Edge / Firefox (latest versions)
- Network Requirements: Stable internet connection for model updates and API sync
- Runtime Environment: Python 3.10+ with virtual environment setup
- Server Type: Localhost during development, scalable to cloud (AWS EC2 / Azure App Service)

Development Environment:

- IDE: Visual Studio Code / PyCharm
- Version Control: Git & GitHub for source code management

- Package Manager: pip for Python dependencies, npm/yarn for frontend
- Documentation: Markdown / README files for technical documentation

Dataset Used

The NLP-Based Resume Analysis and Ranking System utilizes the Resume Screening Dataset published on Hugging Face by AzharAli. This dataset consists of paired resumes and job descriptions, each annotated with relevance **scores** indicating how well a candidate's profile aligns with a particular job post. It contains both structured fields (such as candidate name, experience, and skills) and unstructured text fields (such as job descriptions and summaries), making it suitable for NLP-based similarity and ranking tasks.

The dataset is divided into *train* and *test* splits and is provided in JSON/CSV format, ensuring compatibility with frameworks like pandas, scikit-learn, and PyTorch. Before model training and evaluation, the textual data undergoes cleaning, tokenization, stopwords removal, and vectorization using TF-IDF and Sentence Transformers (MiniLM).

This dataset enables accurate benchmarking for modules such as keyword-based matching, semantic similarity computation using FAISS, and candidate ranking—providing a real-world foundation for resume–job relevance evaluation.

Machine Learning & NLP Libraries

Transformers & Embeddings

- sentence-transformers (all-MiniLM-L6-v2) – for generating semantic embeddings and contextual similarity
- faiss – for fast nearest-neighbour search and vector similarity retrieval
- transformers (Hugging Face) – for pre-trained transformer models such as BART and JobBERT
- torch (PyTorch) – as the backend for deep learning models (LSTM, transformers)

Text Processing & Skill Detection

- scikit-learn – for TF-IDF vectorization, cosine similarity, and preprocessing utilities
- facebook/bart-large-mnli – for zero-shot skill entailment detection (NLI-based inference)
- jjzha/jobbert_skill_extraction – for named entity recognition (skills, degrees, companies)

- re & nltk – for tokenization, regex-based entity cleanup, and text normalization

Document Handling & Visualization

- PyPDF2 / pdfplumber – for reading and extracting resume content from PDF files
- matplotlib / recharts (frontend) – for plotting analytics such as ATS score distribution, skill gap charts, etc.
- pandas & numpy – for data manipulation and analysis during processing

Chapter 7

Project Planning

The development of the NLP-Based Resume Analysis and Ranking System was carried out through a structured, multi-phase approach to ensure systematic progress from conceptualization to deployment. The project began with a requirement analysis and feasibility study, where existing recruitment challenges were examined, and system requirements were defined for both students and HR users. This phase also evaluated the feasibility of implementing NLP and deep learning models locally to maintain data privacy and real-time performance.

In the system design and planning phase, the overall architecture, database schema, and user interface layouts were created. Flow diagrams, block diagrams, and entity-relationship models were prepared to visualize how data would move between the frontend, backend, and NLP modules. The design also specified the integration of models like TF-IDF, Sentence Transformers, BART, FAISS, and LSTM within the FastAPI backend.

The data collection and preprocessing phase focused on gathering resumes and job descriptions, cleaning and structuring the text data, and preparing a skill ontology for zero-shot skill matching. Tools like PyPDF2, Pandas, and NLTK were used for text extraction and preprocessing. Following this, in the model development and training phase, various NLP and deep learning models were implemented and fine-tuned — including TF-IDF with cosine similarity for keyword relevance, Sentence Transformers (MiniLM) for semantic similarity, FAISS for fast vector search, BART for skill entailment detection, JobBERT for entity extraction, and an LSTM model for predicting career progression.

After the models were finalized, the backend and frontend integration phase combined all modules into a unified system. The backend, built using FastAPI and SQLAlchemy, handled authentication, file uploads, and resume analysis, while the React-based frontend (with Chakra UI and Recharts) provided interactive dashboards and analytics for both students and HR. This phase also included role-based access control, ensuring data security and privacy through JWT authentication.

In the testing and evaluation phase, the system underwent extensive validation using metrics such as accuracy, recall, precision, F1-score, and processing time. Integration testing and user acceptance testing were performed to ensure that both student and HR functionalities operated

as intended. Finally, the deployment and documentation phase included packaging the project using Docker, configuring Uvicorn for deployment, and preparing detailed documentation, including the final project report, user manual, and presentation materials.

Overall, the project followed an 11-week development timeline, progressing from requirement gathering to deployment, ensuring each stage contributed to building a robust, efficient, and intelligent recruitment automation system powered by NLP and deep learning.

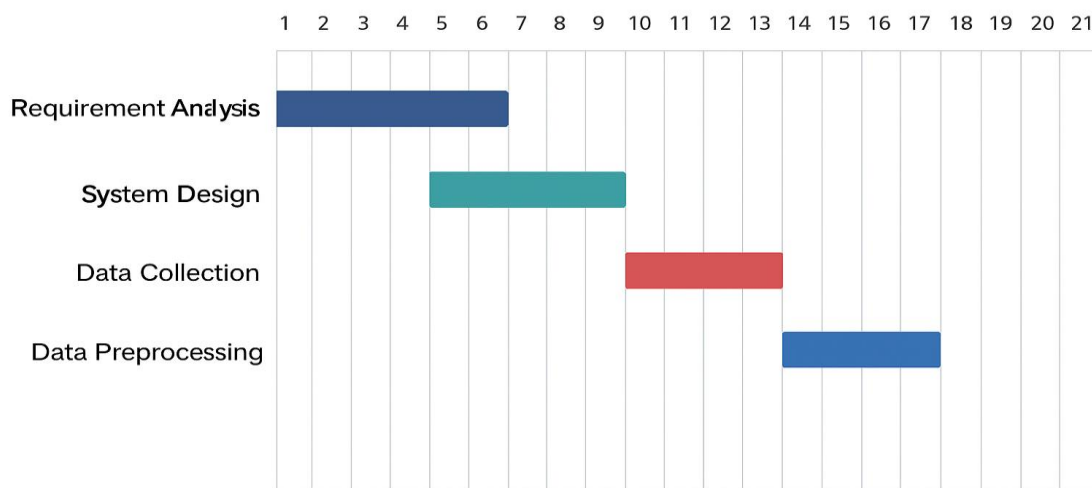


Figure 6.1 Gantt Chart for August

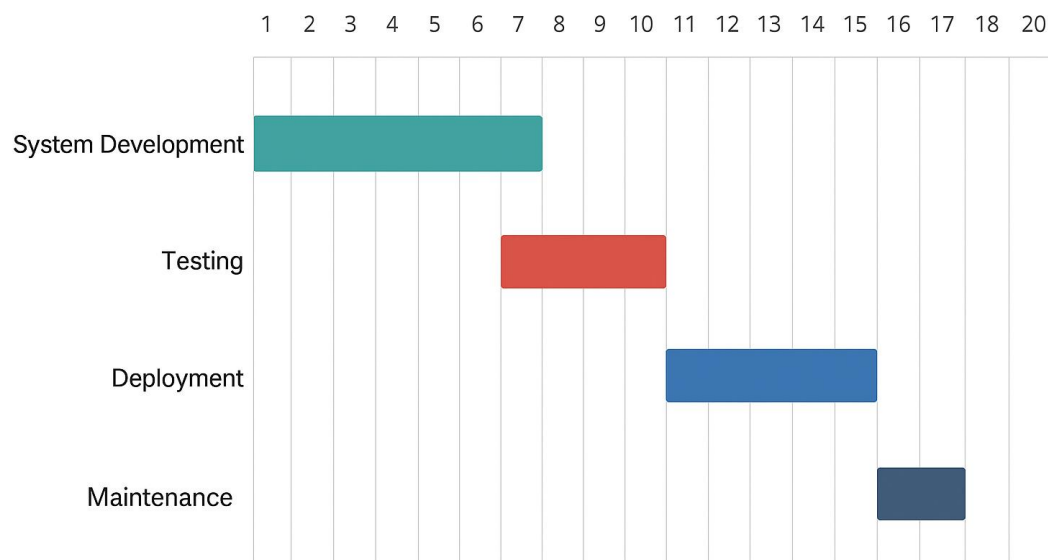


Figure 6.2 Gantt Chart for September

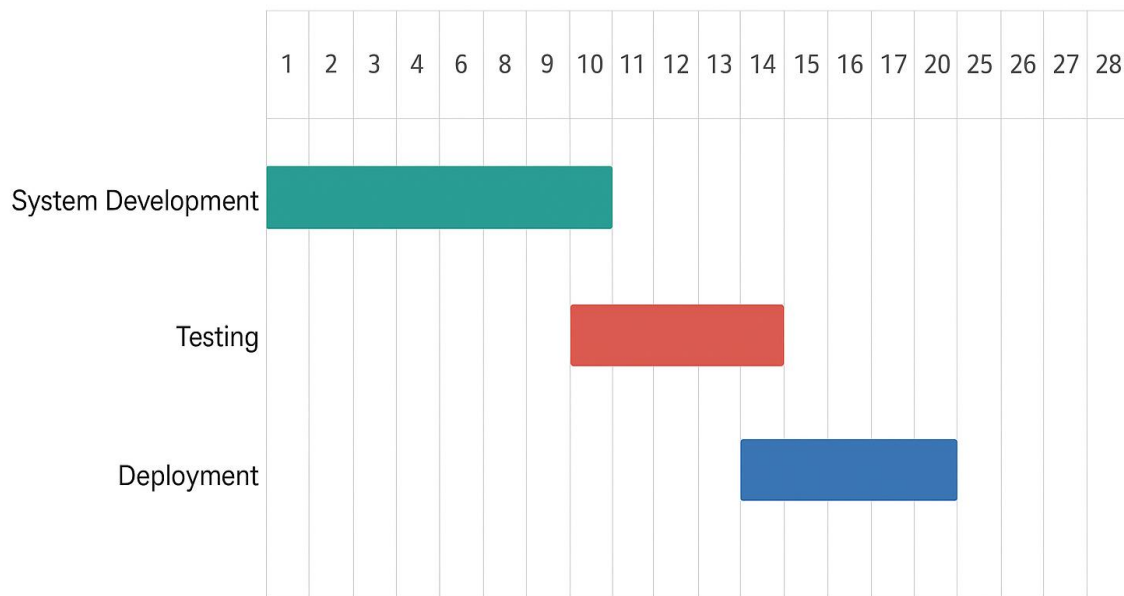


Figure 6.3 Gantt Chart for October & November

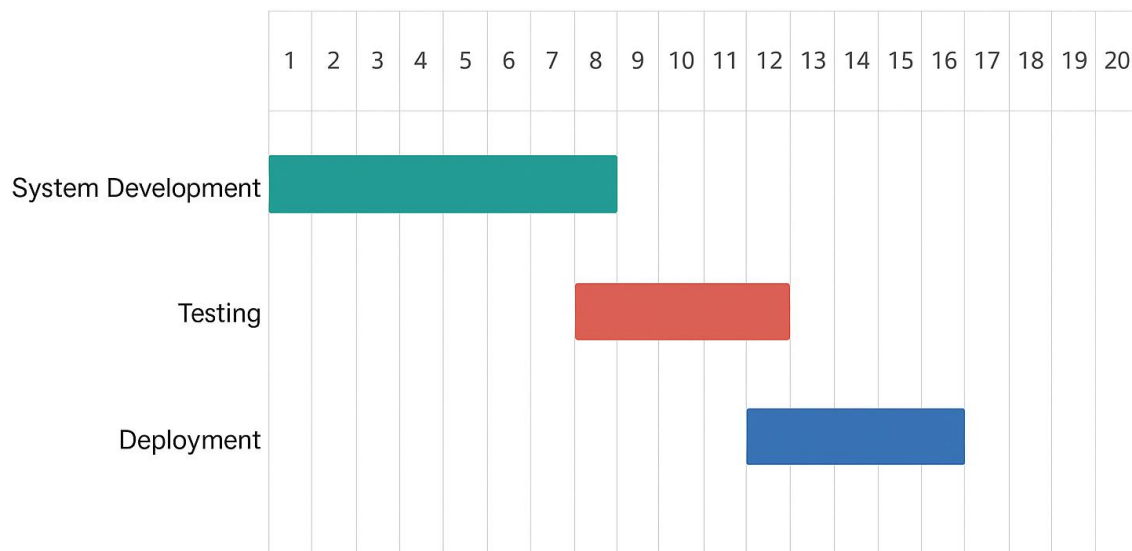


Figure 6.4 Gantt Chart for November

Chapter 8

Expected Outcome

The NLP-Based Resume Analysis and Ranking System is designed to produce an intelligent, automated, and explainable recruitment support platform that benefits both students/job seekers and HR professionals. The system integrates NLP, ML, and DL techniques to deliver the following expected outcomes:

1. For Students / Job Seekers:

- **Automated Resume Analysis:** The system will analyse uploaded resumes against a selected job description and generate an ATS score, success probability, and skill gap analysis.
- **Personalized Feedback:** Students receive detailed insights such as strengths, areas of improvement, and resume optimization tips based on NLP-driven evaluation.
- **Skill Gap Recommendations:** The platform recommends courses and learning paths to help users bridge missing or weak skills.
- **Mock Interview Support:** Students can access mock interview modules tailored to their target role to improve confidence and performance.
- **Application Tracking:** Users can track job applications and manage their history with progress visualization and insights.
- **Resume Enhancement:** Students can refine their resumes according to feedback and download updated, optimized versions automatically.

2. For HR Professionals / Recruiters:

- **Job Posting Management:** HR users can create, edit, and manage job listings, specifying skills, experience levels, and salary ranges.
- **Bulk Resume Upload and Analysis:** HR professionals can upload multiple resumes at once and receive AI-driven analysis with ATS scoring, keyword matching, and candidate ranking.
- **Candidate Ranking Dashboard:** The system provides a ranked list of candidates with detailed reasoning including semantic similarity, experience match, skills match, and education fit.

- **Profile Management:** HRs can update their professional profiles, company information, and manage recruitment activities effectively.
- **Analytics and Insights:** HRs can view visual analytics such as candidate pipeline, hiring trends, and skill distribution using dashboards built with Recharts and Chakra UI.

3. System-Level Outcomes:

- **Efficiency and Accuracy:** Automated analysis reduces manual screening time and improves consistency in candidate evaluation.
- **Data Privacy and Local Inference:** All NLP and DL models operate locally (using Sentence Transformers, BART, JobBERT, LSTM, and FAISS), ensuring secure, cost-effective, and real-time processing.
- **Explainable AI:** Transparent scoring and reasoning ensure that both students and HRs understand the evaluation process.
- **Scalability and Integration:** The modular FastAPI architecture and React-based frontend make the system scalable for future enhancements such as chatbot integration, predictive analytics, and multilingual support.

Implementation

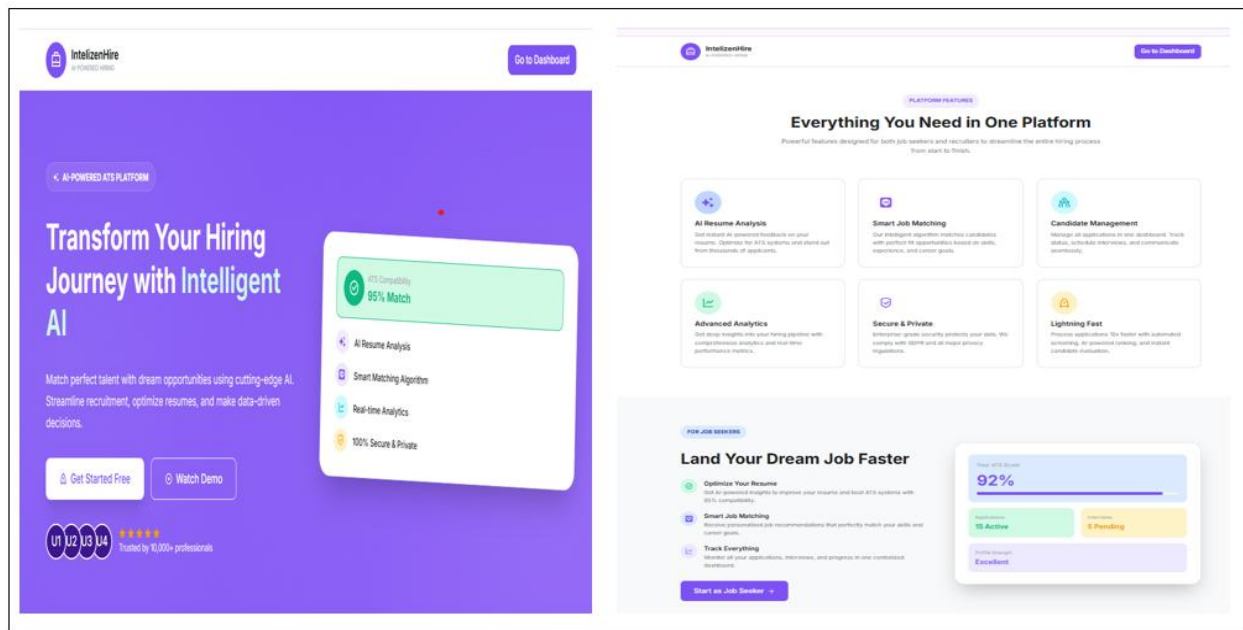


Figure 5.4 Landing Page

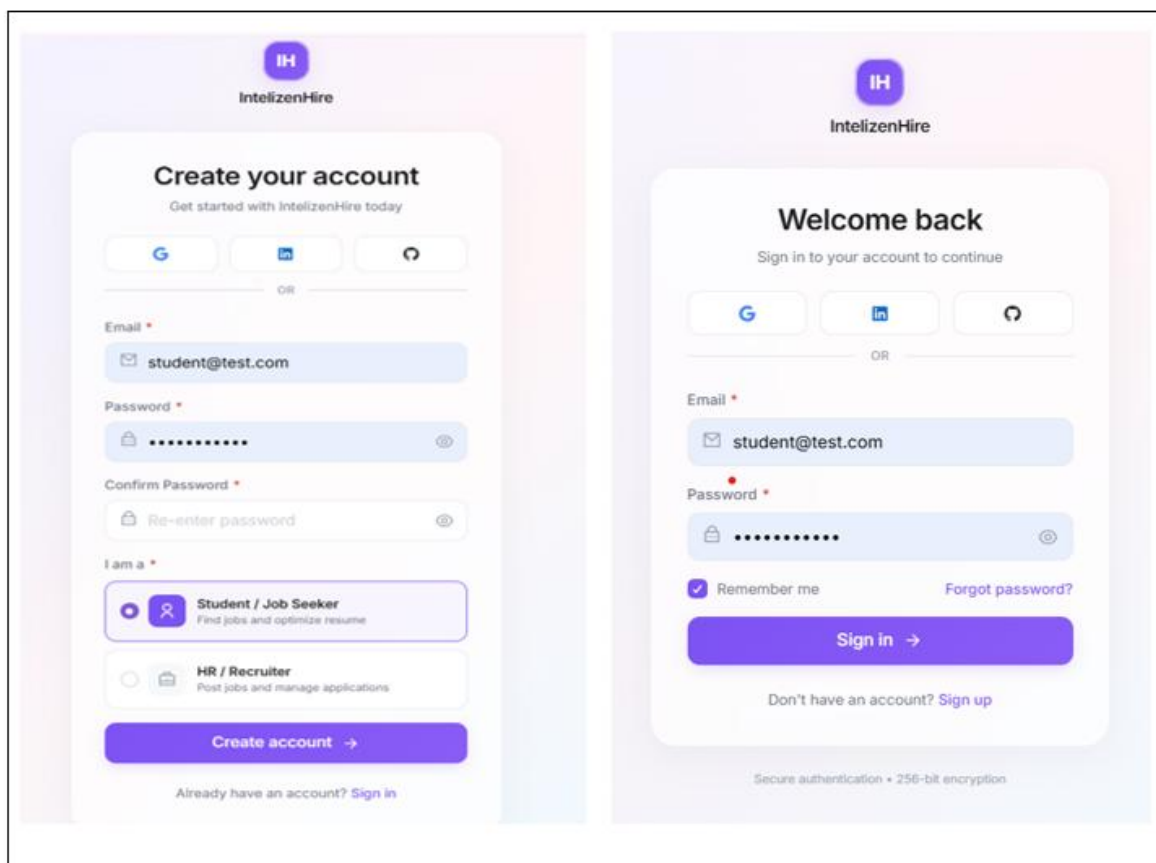


Figure 5.5 Register and Sign in Page

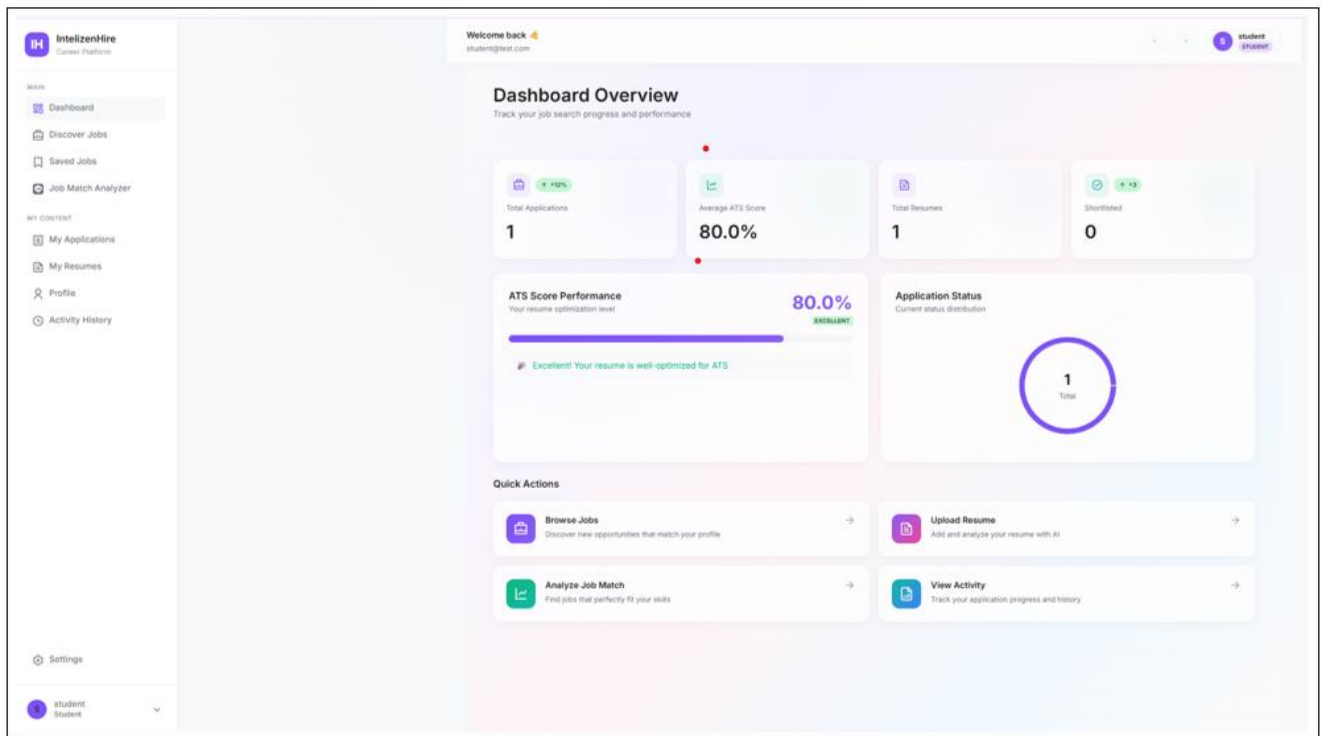


Figure 5.6 Student Dashboard

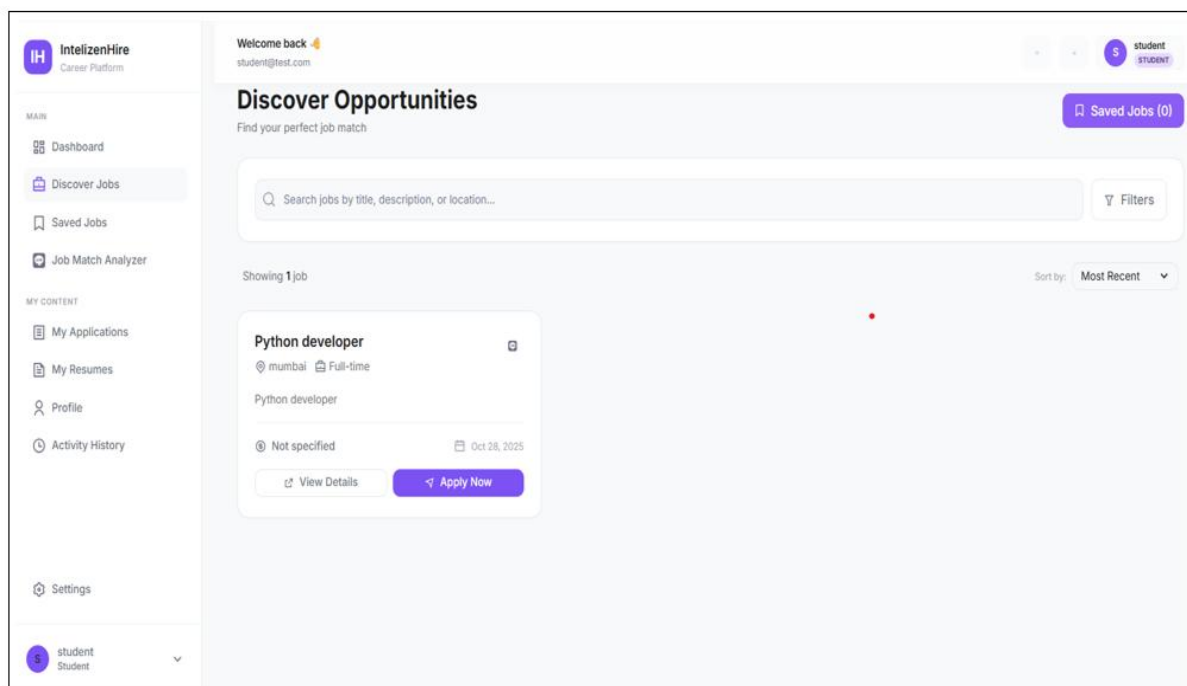


Figure 5.7 Student Job Browsing

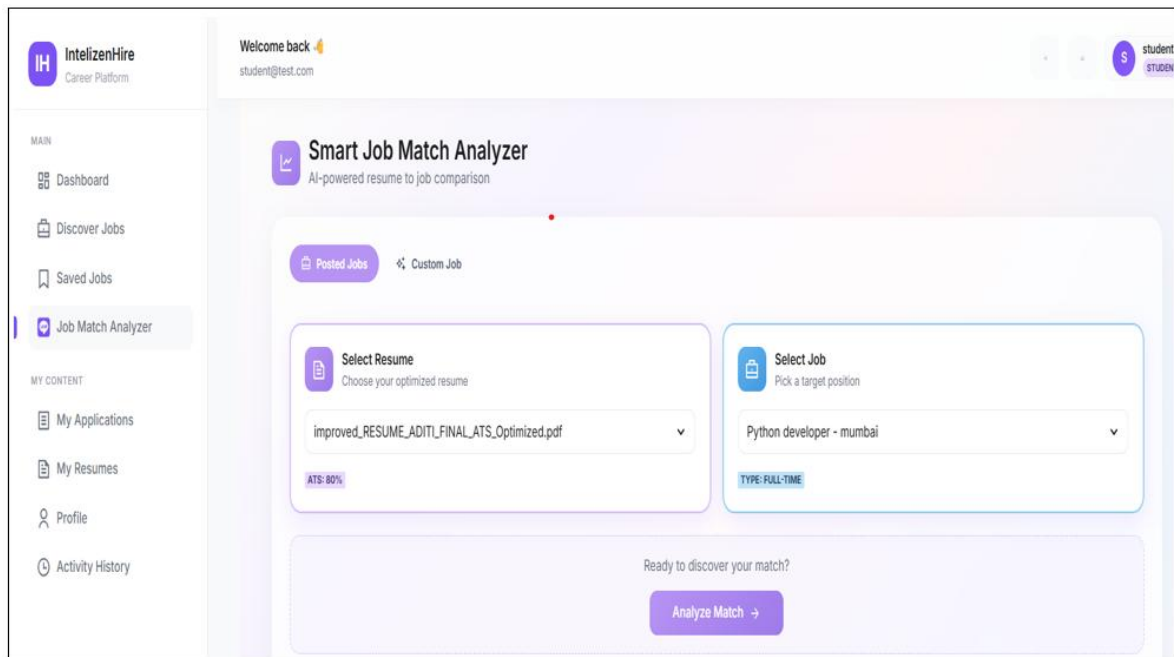


Figure 5.8 Student Resume & Job Matcher

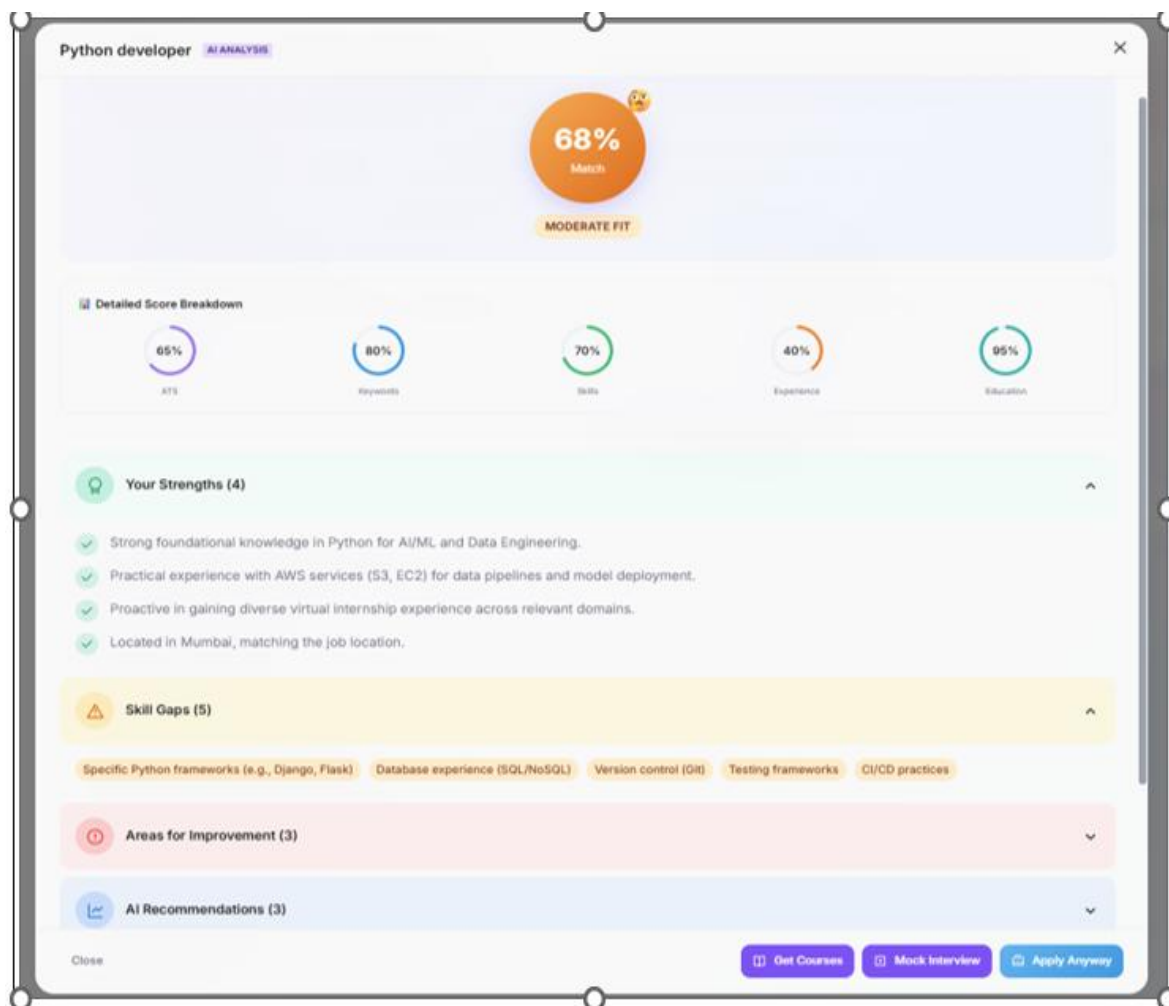


Figure 5.9 Student Resume Analysis

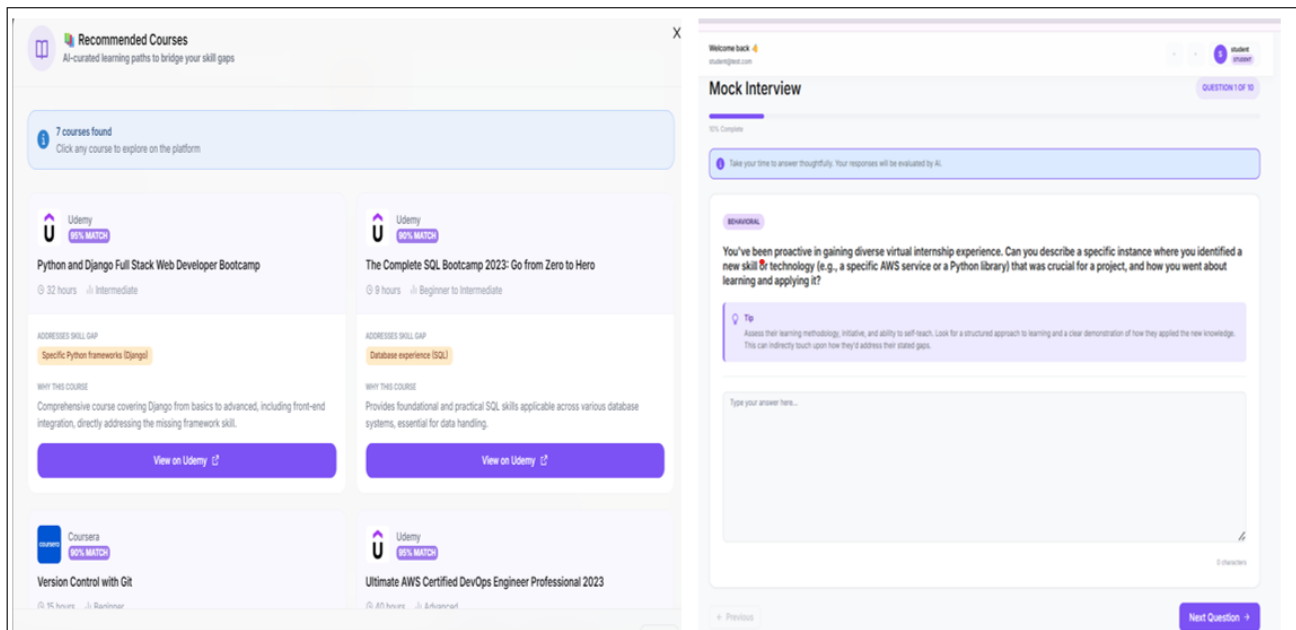


Figure 5.10 Student: Recommendation Courses & Mock Interviews

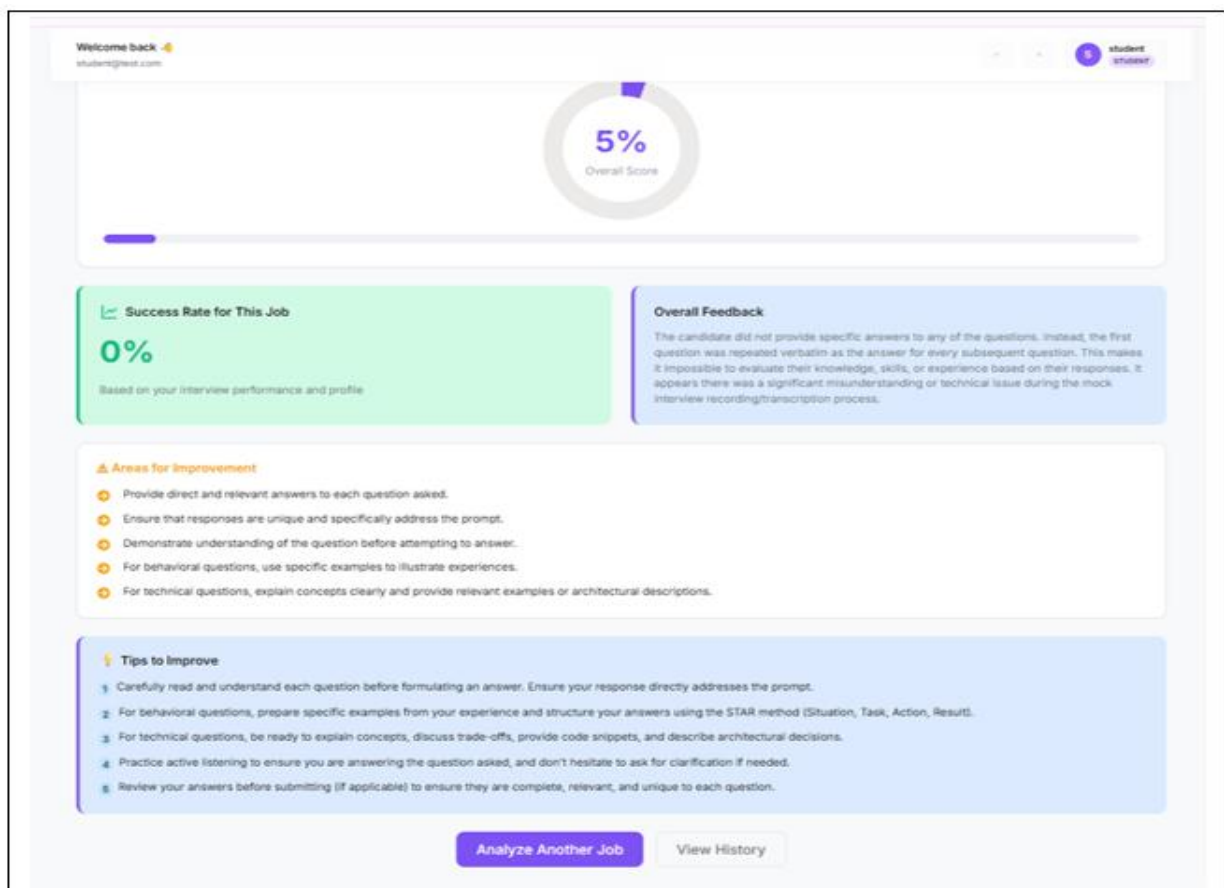


Figure 5.11 Job Matching Analysis

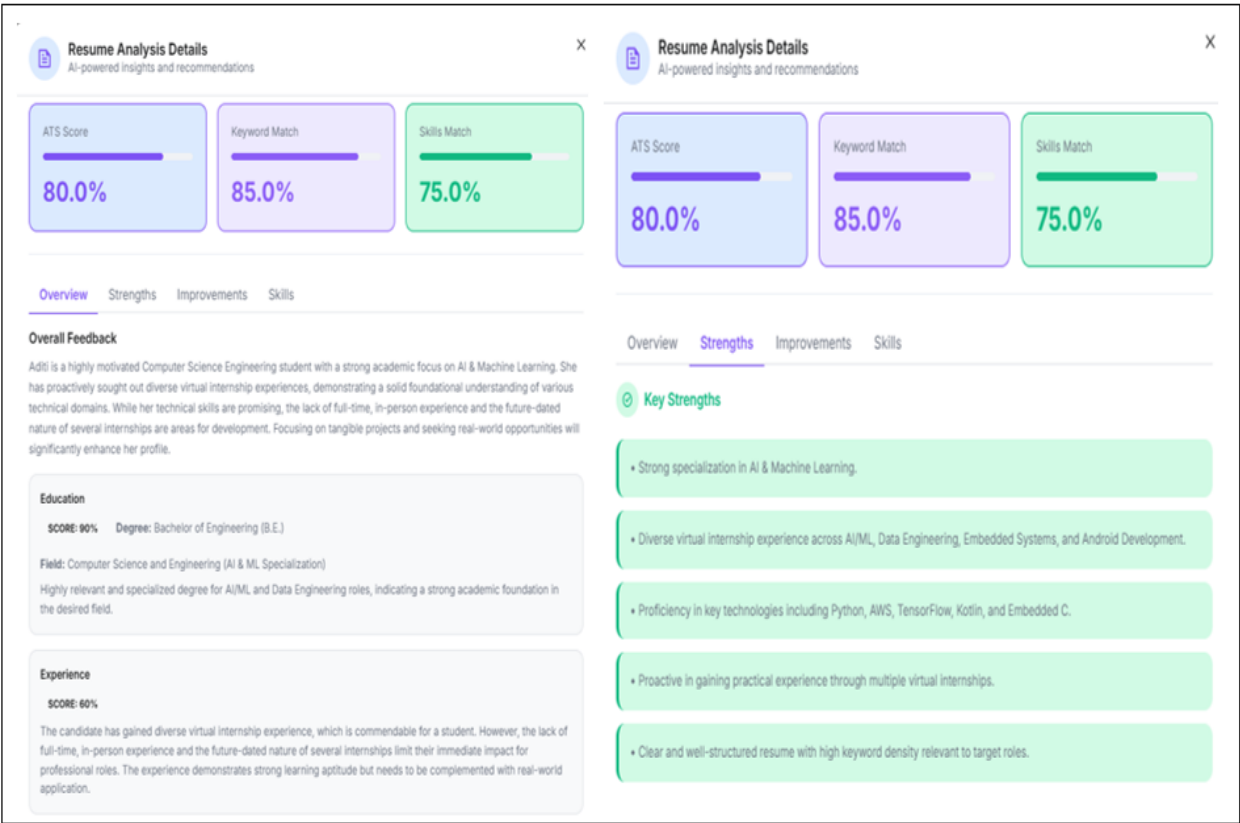


Figure 5.12 Student Resume Analysis (Overview & Strengths)

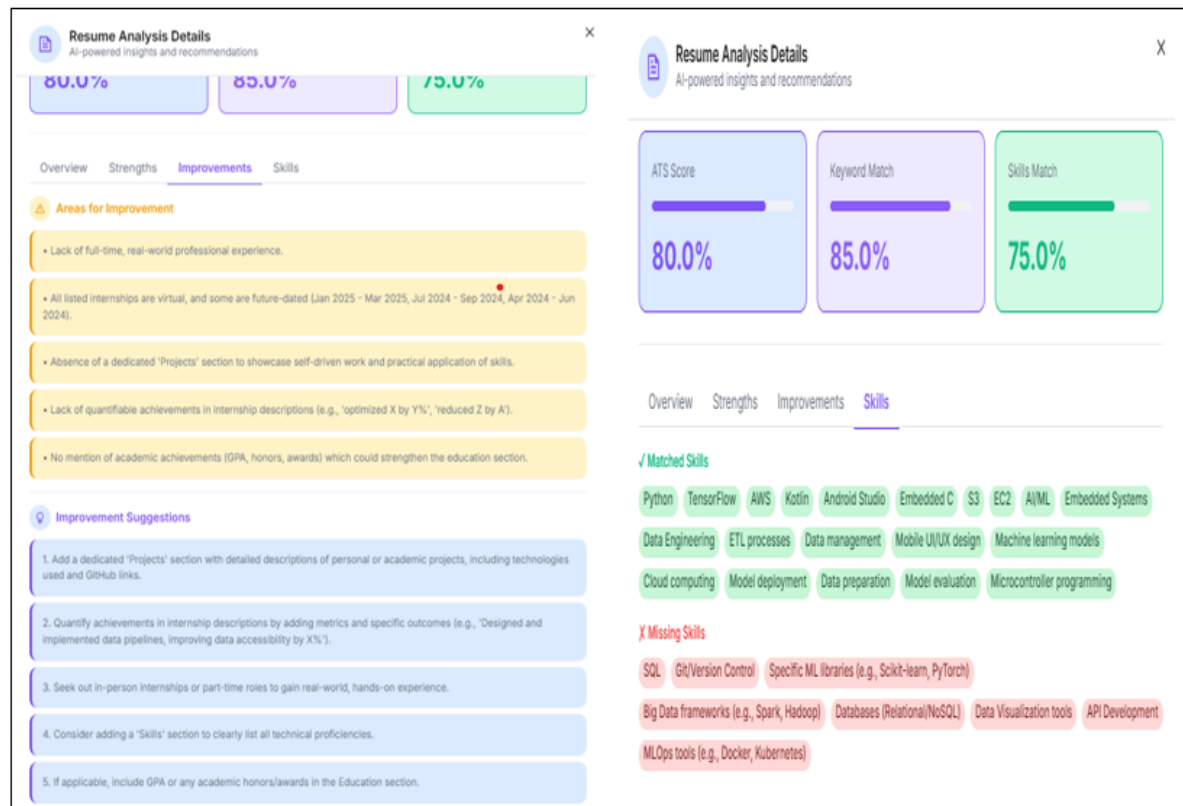


Figure 5.13 Student Resume Analysis (Improvements & Skills)

IntelizenHire

Career Platform

Welcome back 🌟

student@test.com

student

STUDENT

MAIN

- Dashboard
- Discover Jobs
- Saved Jobs
- Job Match Analyzer

MY CONTENT

- My Applications
- My Resumes
- Profile
- Activity History

Settings

student Student

My Profile

Personal Info

Education

Experience

<> Skills

Personal Information

Basic details about yourself

First Name *

Aditi

Last Name *

Gadhav

Phone Number

+1 (555) 000-0000

Date of Birth

dd-mm-yyyy

Location

San Francisco, CA

Professional Links

Showcase your work and connect

LinkedIn Profile

https://linkedin.com/in/johndoe

GitHub Profile

Portfolio Website

Figure 5.14 Student Profile Updation

IntelizenHire

Career Platform

Welcome back 🌟

student@test.com

student

STUDENT

MAIN

- Dashboard
- Discover Jobs
- Saved Jobs
- Job Match Analyzer

MY CONTENT

- My Applications
- My Resumes
- Profile
- Activity History

Activity History & Analytics

Track your progress and view detailed insights

1 Total Resumes

1 Applications

4 Job Comparisons

80.0% Avg ATS Score

Resumes (1)

Applications (1)

Comparisons (4)

Interviews (2)

FILE NAME	UPLOAD DATE	ATS SCORE	ANALYSES
improved_RESUME_ADITI_FINAL_ATS_Optimized.pdf	Oct 28, 2025	80.0%	4 times

Application Status Breakdown

1 Submitted

Figure 5.15 Student Activity History

33

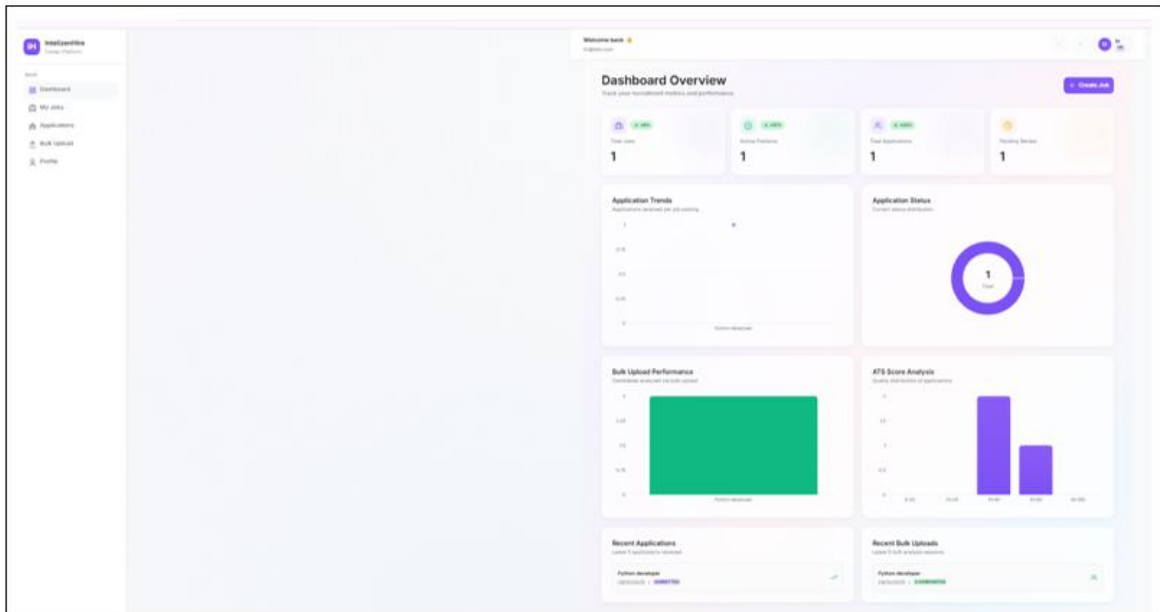


Figure 5.16 HR Dashboard

The screenshot shows the 'Create New Job' form in the IntelzenHire system. The form is titled 'Create New Job' and includes a sub-header 'Post a new position to attract candidates'. The form is divided into several sections:

- Basic Information:** Includes fields for Job Title (Senior Software Engineer), Department (Engineering), Location (San Francisco, CA), Employment Type (Full Time), Experience Level (Mid Level), Minimum Salary (80000), and Maximum Salary (120000).
- Job Description:** A text area for describing the job role.
- Requirements:** A section for adding requirements, with an 'Add requirement' button.
- Responsibilities:** A section for adding responsibilities, with an 'Add responsibility' button.
- Required Skills:** A section for adding skills, with an 'Add skill' button.
- Benefits:** A section for adding benefits, with an 'Add benefit' button.
- Publication Options:** A checkbox for 'Publish job immediately (make visible to candidates)'.
- Create Job as Draft:** A large blue button at the bottom.

Figure 5.17 HR Job Posting

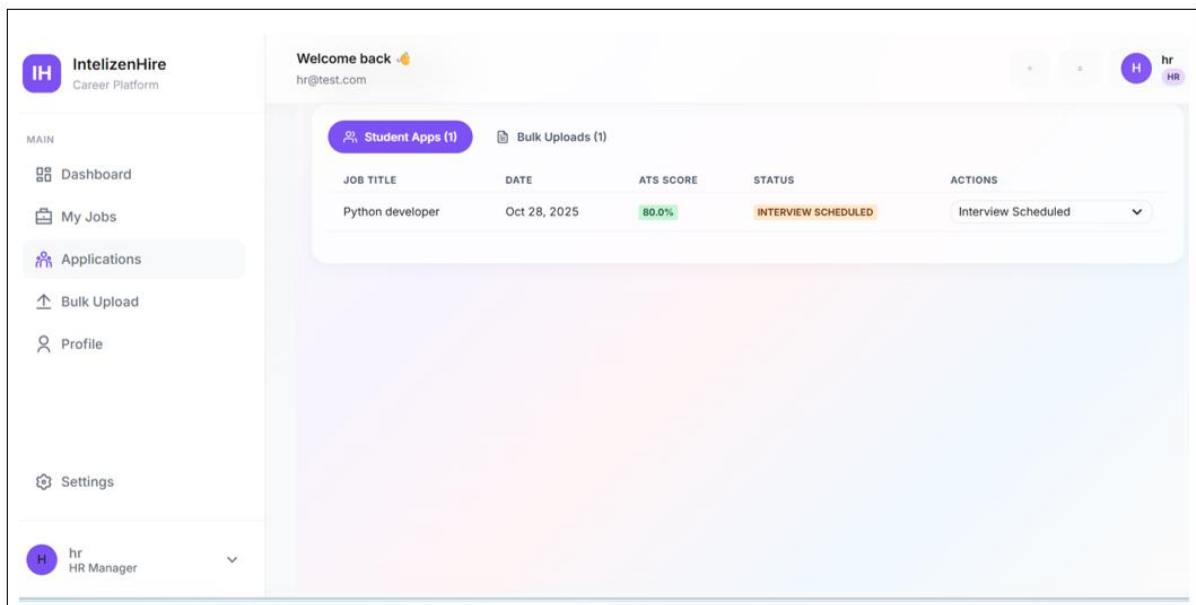


Figure 5.18 HR Application Manager

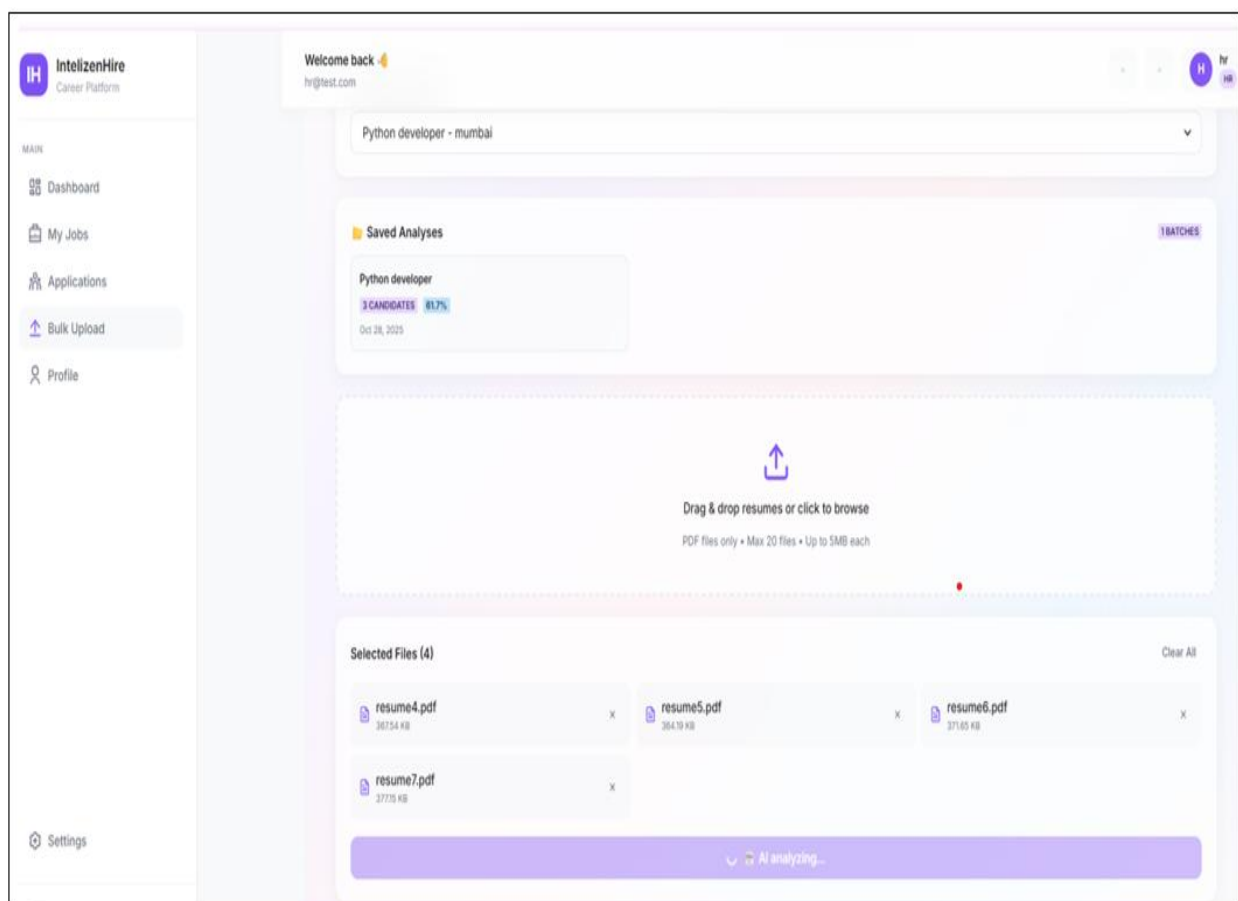


Figure 5.19 HR Bulk Resume Upload

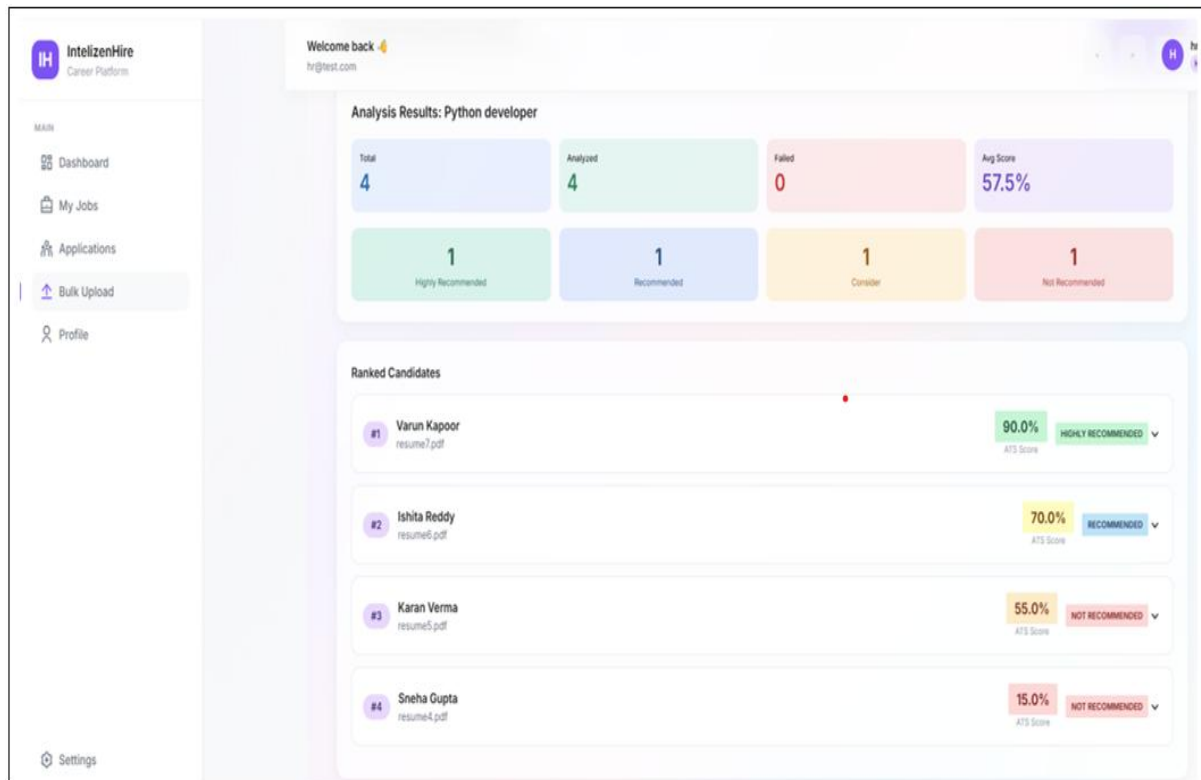


Figure 5.20 HR Candidate Ranking

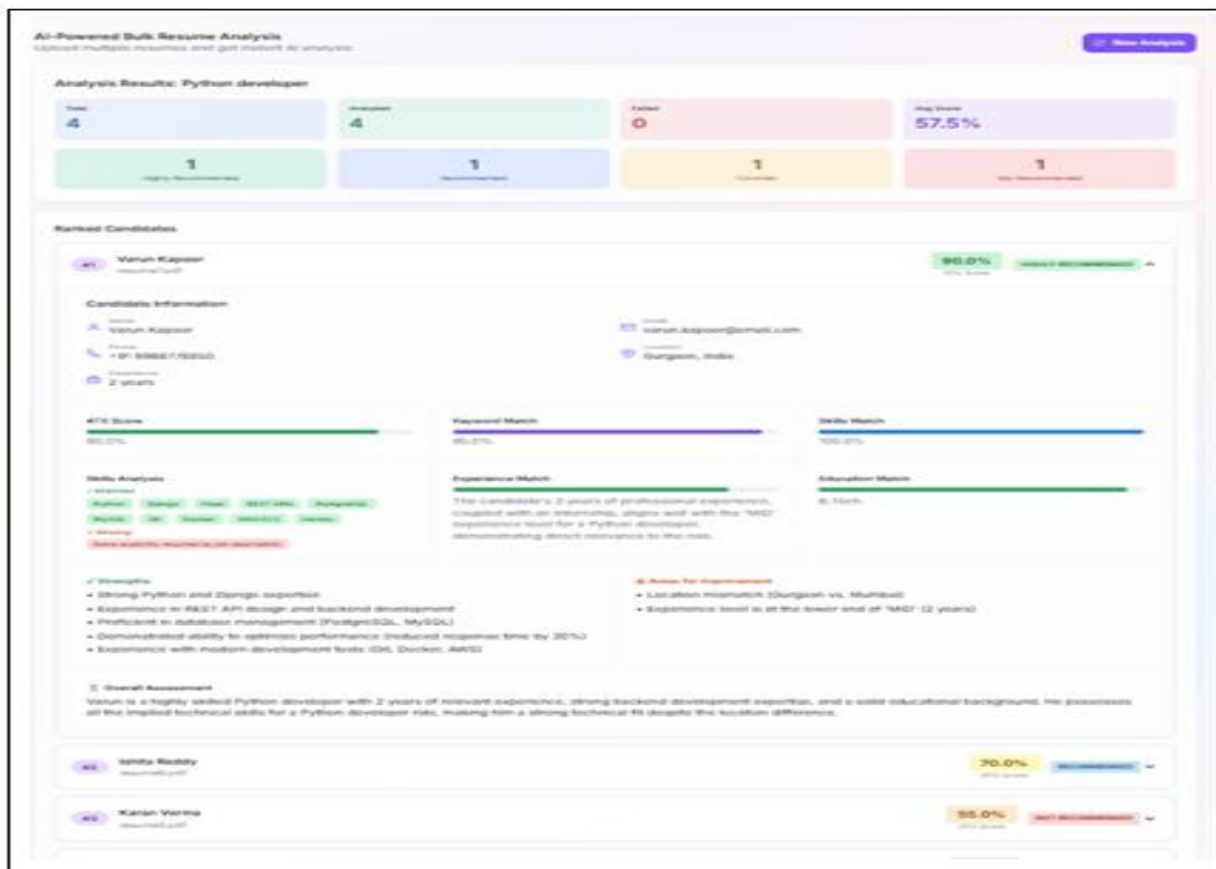


Figure 5.21 HR Candidate Ranking Analysis

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